We now need to select those itineraries that have all the cities in them. (Remember we have DNA molecules that are five cities long, but in which cities may be repeated or missing). Now to pick out the molecules that contain a specific city, we use a process called affinity purification. We do this using, of all things, magnetic beads: the complement of the sequence you're interested in is attached to a magnetic bead. This complement hybridises with the sequence you're after. (Again, we're using the hybridising property of DNA.) Then you simply retrieve the bead, and you're left with DNA that *does* contain the sequence you're after.

It should be patent that you need to do affinity purification five times. For example, Mumbai *must* be in the pool of candidates; so you attach the complement of Mumbai to the magnetic beads, it hybridises with all the molecules that have Mumbai in them, and you pull out all the beads. The ones that get left behind are the sequences that *don't* have Mumbai in them. Then, in the next test tube, you affinity-purify for Delhi, and so on. In the last test tube, you'll be left with itineraries that visit each city exactly once, and which start at Mumbai and end at Kolkata!

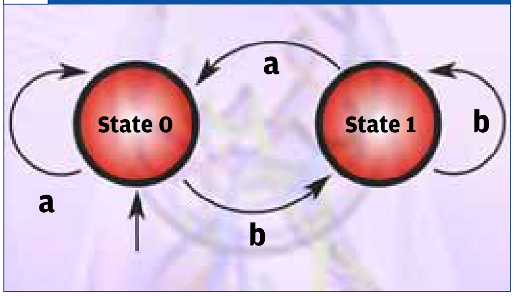
Here, we 'know' the answer: it should be Mumbai-Delhi-Bangalore-Chennai-Kolkata. So we need to find if our 'solution' test tube has the correct path. The technique used is PCR again, combined with the technique for measuring the length of a DNA sequence, called gel electrophoresis, which we've used before. You amplify (using PCR) the section between Mumbai and, say, Bangalore. If these molecules are 12 base pairs long (as they should be if all went well), you know Bangalore is the third city in the itinerary! Similarly, you PCR-amplify for all the four remaining cities and find their positions in the itinerary. We're done!

Of course, in our example, we haven't really solved a problem, but we've shown (like Adleman did) that DNA and the procedures mentioned above can be used to solve a mathematical problem. Unfortunately, even what Adleman did has not convinced everyone. Calculations have been done to show that if the problem were to be increased to a 100 or so cities, the amount of DNA required would be utterly impossible; here, it's a restriction in terms of volume, rather than a restriction in terms of time as with a traditional computer.

So Is DNA Computing Any Good?

Yes. It's only that we haven't gotten there yet. One must realise that DNA computing is, at the current state of the art, best applicable to problems that are massively parallel in nature. This includes problems of the sort we described above. Then, there's encryption, where pretty much the same thing applies: to crack a code, one needs to try a whole lot of possibilities at the same time. There also happen to be several computer science issues that can be explored better by a DNA-based computing system, by virtue of its "stochastic" (based on probability and statistics) nature.

A Finite Automaton



A simple two-state, two-symbol finite automaton (FA). The FA is a 'machine' that has a state. When it's in state 0 and the input is b, the machine goes to state 1. When it's in state 1 and the input is b, it remains in state 1, and so on

At the same time, advances in biotechnology are about as staggering as those happening in the silicon world. We're just not aware of them! We've mentioned techniques, involving enzymes and magnetic beads, in our sample 'problem': these would have been impossible to do in, say, 1970.

There's a lot of the word 'self' associated with DNA. DNA molecules self-replicate; in so doing, they self-refer, with proteins as the intermediary; and it turns out that DNA's biggest promise lies in its ability to self-assemble. Much research in recent years has focused on this property of the DNA molecule—rather, of the structures that can be built at the nanoscale from the DNA molecule itself.

So what is self-assembly? And what is the relation between self-assembly and computation? These are important questions, because we're asking if DNA computing is just a fad or if it is the future. It turns out that DNA, in its self-assembling nature, could be a universal computer. In other words, it might be used someday to build any kind of computer at all!

Now that's a possible happy ending. But if you'd like the gory details, read on...

"A Trillion Computers In A Drop Of Water"

Such headlines make for exciting news. This one was in November 2001, when several Web sites reported that a group of scientists led by Prof Ehud Shapiro at the Weizmann Institute of Science in Israel had created a programmable FA (finite automaton; see figure) in a test tube. It was a two-state, two symbol FA, which is not very complex as FAs go, but it *is* an achievement. The news went: "This nanocomputer is so small that a trillion such computers compute in parallel, in a drop the size of a tenth of a millilitre. Collectively, the computers perform a billion operations per second...